Advanced Trigonometry

ANSWER KEY

Section 1: Angle Sum, Difference & Double-Angle Formulas

1 If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant II, what is the exact value of $\sin(2\theta)$?

a. $-\frac{24}{25}$ ✓

b. $\frac{24}{25}$

c. $-\frac{7}{25}$

d. $\frac{7}{25}$

In Quadrant II, $\cos\theta = -\sqrt{1-(3/5)^2} = -\frac{4}{5}$. Then $\sin(2\theta) = 2\sin\theta\cos\theta = 2(\frac{3}{5})(-\frac{4}{5}) = -\frac{24}{25}$.

2 What is the exact value of $\cos(75^\circ)$ using the angle sum formula $\cos(45^\circ + 30^\circ)$?

a. $\frac{\sqrt{6}-\sqrt{2}}{4}$ ✓

b. $\frac{\sqrt{6}+\sqrt{2}}{4}$

c. $\frac{\sqrt{2}-\sqrt{6}}{4}$

d. $\frac{\sqrt{3}-1}{2}$

$\cos(45^\circ + 30^\circ) = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ = (\frac{\sqrt{2}}{2})(\frac{\sqrt{3}}{2}) - (\frac{\sqrt{2}}{2})(\frac{1}{2}) = \frac{\sqrt{6}-\sqrt{2}}{4}$.

3 Which of the following is equivalent to $\frac{1-\cos(2x)}{2}$?

a. $\sin^2 x$ ✓

b. $\cos^2 x$

c. $\tan^2 x$

d. $\sin(2x)$

This is the standard power-reducing identity: $\sin^2 x = \frac{1-\cos(2x)}{2}$.

Section 2: Solving Trigonometric Equations

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Solve the equation $2\sin^2 x - 3\sin x + 1 = 0$ for all solutions in the interval $[0, 2\pi)$.

a. $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$ ✓

b. $\frac{\pi}{3}, \frac{2\pi}{3}, \pi$

c. $\frac{\pi}{6}, \frac{7\pi}{6}$

d. $\frac{\pi}{2}, \frac{3\pi}{2}$

Factor: $(2\sin x - 1)(\sin x - 1) = 0$. Thus $\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$, or $\sin x = 1 \Rightarrow x = \frac{\pi}{2}$.

5 How many solutions does the equation $\cos(2x) = 0$ have in the closed interval $[0, 2\pi]$?

a. 2

b. 4 ✓

c. 6

d. 8

For $x \in [0, 2\pi]$, $2x \in [0, 4\pi]$. $\cos(2x) = 0$ when $2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$. Dividing by 2 yields 4 distinct solutions.

6 Solve for $x \in [0, \pi]$: $\sqrt{3}\tan(2x) = 1$.

a. $\frac{\pi}{12}, \frac{7\pi}{12}$ ✓

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{3}$

d. $\frac{\pi}{4}, \frac{3\pi}{4}$

$\tan(2x) = \frac{1}{\sqrt{3}}$. For $2x \in [0, 2\pi]$, $2x = \frac{\pi}{6}$ or $2x = \frac{7\pi}{6} \Rightarrow x = \frac{\pi}{12}$ or $x = \frac{7\pi}{12}$.

Section 3: Inverse Trigonometric Functions

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 What is the exact value of $\sin(\arccos(-\frac{1}{2}))$?

a. $\frac{\sqrt{3}}{2}$ ✓

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{1}{2}$

d. $-\frac{1}{2}$

$\arccos(-\frac{1}{2}) = \frac{2\pi}{3}$. Then $\sin(\frac{2\pi}{3}) = \frac{\sqrt{3}}{2}$.

8 Evaluate $\tan(\arcsin(\frac{5}{13}))$.

a. $\frac{5}{12}$ ✓

b. $\frac{12}{5}$

c. $\frac{5}{13}$

d. $\frac{12}{13}$

Let $\theta = \arcsin(5/13)$, so opposite $= 5$, hypotenuse $= 13$, adjacent $= \sqrt{13^2 - 5^2} = 12$. Therefore, $\tan\theta = \frac{5}{12}$.

9 What is the range of the principal inverse tangent function $f(x) = \arctan(x)$?

a. $(-\frac{\pi}{2}, \frac{\pi}{2})$ ✓

b. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

c. $[0, \pi]$

d. $(0, \pi)$

The standard range of $\arctan(x)$ is the open interval $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Section 4: Law of Tangents & Polar Trigonometry

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 Which identity represents $\cos(A - B)$?

a. $\cos A\cos B + \sin A\sin B$ ✓

b. $\cos A\cos B - \sin A\sin B$

c. $\sin A\cos B - \cos A\sin B$

d. $\sin A\cos B + \cos A\sin B$

By standard cosine subtraction theorem: $\cos(A - B) = \cos A\cos B + \sin A\sin B$.

11 In $\triangle ABC$, $a = 7$, $b = 5$, and $c = 8$. What is $\cos B$?

a. $\frac{11}{14}$ ✓

b. $\frac{1}{2}$

c. $\frac{5}{7}$

d. $\frac{13}{16}$

Law of Cosines: $\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{49 + 64 - 25}{2(7)(8)} = \frac{88}{112} = \frac{11}{14}$.

12 What is the maximum value of the function $f(x) = 3\sin x - 4\cos x$?

a. 1

b. 5 ✓

c. 7

d. $\sqrt{7}$

Using the linear combination harmonic form $A\sin x + B\cos x = R\sin(x + \phi)$, the amplitude is $R = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647